

R Lab 2. Review of T-tests and F-tests

Set a working directory. Yours is different from mine. It's where you saved the data file from Blackboard.

```
> setwd("C:\\Users\\baron\\Documents\\Teach\\615 Regression\\Data")
> H = read.csv("HOME_SALES.csv")
> attach(H)
> head(H)
```

	ID	SALES_PRICE	FINISHED_AREA	BEDROOMS	BATHROOMS	GARAGE_SIZE	YEAR_BUILT	STYLE
1	1	360.0	3032	4	4	2	1972	1
2	2	340.0	2058	4	2	2	1976	1
3	3	250.0	1780	4	3	2	1980	1
4	4	205.5	1638	4	2	2	1963	1
5	5	275.5	2196	4	3	2	1968	3
6	6	248.0	1966	4	3	5	1972	1

	LOT_SIZE	AIR_CONDITIONER	POOL	QUALITY	HIGHWAY
1	22221	YES	NO	MEDIUM	NO
2	22912	YES	NO	MEDIUM	NO
3	21345	YES	NO	MEDIUM	NO
4	17342	YES	NO	MEDIUM	NO
5	21786	YES	NO	MEDIUM	NO
6	18902	YES	YES	MEDIUM	NO

No need to print all 522 rows of data. To get an idea, "head" is a good command, showing the first few lines only.

1. A one-sample T-test

1a. A one-sample, two-sided T-test

There is a claim that the average price of homes in the region is \$300,000. Does the data set support or disprove the claim? This is a two-sided test because there is no specified direction, we are just testing if the population mean is 300,000 or not.

```
> t.test(SALES_PRICE, mu=300)
```

One Sample t-test

```
data: SALES_PRICE
t = -3.6619, df = 521, p-value = 0.0002759
alternative hypothesis: true mean is not equal to 300
95 percent confidence interval:
 266.0348 289.7535
sample estimates:
mean of x
 277.8941
```

Conclusion: the p-value is very low, hence, there is significant evidence that the mean home price is not \$300,000. We also find that the sample mean price in the data set is \$277,894, the observed t-statistic is $t = -3.66$, and the 95% confidence interval for the mean price is [\$266,035, \$289,754]. You may recall the duality between hypothesis testing and confidence estimation: the level α two-sided test rejects the null hypothesis if and only if the $(1 - \alpha)100\%$ confidence interval does not contain the tested parameter value. Here we see that the confidence interval does not contain \$300,000, and no surprise, H_0 is rejected.

To compute the t-statistic by hand, we calculated the sample mean and standard deviation

```
> mean(SALES_PRICE)
```

```
[1] 277.8941
```

```
> sd(SALES_PRICE)
```

```
[1] 137.9234
```

and used the formula for the Student's t-ratio. Or, all in one step,

```
> t = (mean(SALES_PRICE) - 300) / (sd(SALES_PRICE)/sqrt(length(SALES_PRICE)))
```

```
> t
```

```
[1] -3.661884
```

1b. A one-sample, left-tail T-test.

Is the mean price less than \$300,000? This is a one-sided, left-tail test.

```
> t.test(SALES_PRICE, mu=300, alternative="less")
```

One Sample t-test

data: SALES_PRICE

$t = -3.6619$, $df = 521$, $p\text{-value} = 0.000138$

alternative hypothesis: true mean is less than 300

95 percent confidence interval:

$-\text{Inf}$ 287.8414

sample estimates:

mean of x

277.8941

We noticed and explained in class why this p-value is exactly a half of the two-sided p-value. It's very small, so we conclude that yes, there is a significant evidence that the mean home price is less than \$300,000.

1c. A one-sample, right-tail T-test.

Is there any evidence that the mean price is *above* \$300,000? Now, this is a one-sided, right-tail test.

```
> t.test(SALES_PRICE, mu=300, alternative="greater")
```

One Sample t-test

```

data: SALES_PRICE
t = -3.6619, df = 521, p-value = 0.9999
alternative hypothesis: true mean is greater than 300
95 percent confidence interval:
 267.9469      Inf
sample estimates:
mean of x
 277.8941

```

This p-value is a complement of the previous one, and it is very high. No evidence that the population mean exceeds \$300,000. Certainly! If the sample mean is below 300, it has no way to support a claim that the population mean is above 300.

2. A two-sample T-test

Does the sales price depend on the presence of a pool? To answer this question, we have to compare homes with the pool and without it. This is a comparison of two populations, so it is a two-sample test.

```
> t.test(x=SALES_PRICE[POOL=="YES"], y=SALES_PRICE[POOL=="NO"])
```

Welch Two Sample t-test

```

data: SALES_PRICE[POOL == "YES"] and SALES_PRICE[POOL == "NO"]
t = 3.428, df = 40.546, p-value = 0.001408
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 32.74042 126.70831
sample estimates:
mean of x mean of y
 352.1203  272.3959

```

This test compared the mean of sample X with the mean of sample Y, homes with the pool and without the pool. We find a significant evidence that the mean prices are different in the population, and thus, the price does depend on a pool. The difference between mean prices with and without a pool has 95% confidence limits \$32,730 and \$126,708. Notice a non-integer number of degrees of freedom. It is calculated by the Satterthwaite approximation.

A more elegant way of doing the same test is:

```
> t.test( SALES_PRICE ~ POOL )
```

Welch Two Sample t-test

```

data: SALES_PRICE by POOL
t = -3.428, df = 40.546, p-value = 0.001408
alternative hypothesis: true difference in means between group NO and group YES is not equal to 0
95 percent confidence interval:
 -126.70831 -32.74042
sample estimates:
mean in group NO mean in group YES
  272.3959      352.1203

```

Same result, of course.

```
> t.test( SALES_PRICE ~ POOL, alternative="greater")
```

```
Welch Two Sample t-test
```

```
data: SALES_PRICE by POOL
```

```
t = -3.428, df = 40.546, p-value = 0.9993
```

```
alternative hypothesis: true difference in means between group NO and group YES is greater than 0  
95 percent confidence interval:
```

```
-118.8728      Inf
```

```
sample estimates:
```

```
mean in group NO mean in group YES
```

```
272.3959      352.1203
```

There is significant evidence that homes with the pool are *more expensive*, on the average.

3. A two-sample F-test of variances

This F-test is used to compare variances of two samples and in particular, to decide which two-sample T-test is appropriate – a test that assumes equal variances or the Satterthwaite approximation.

```
> var.test(SALES_PRICE ~ POOL)
```

```
F test to compare two variances
```

```
data: SALES_PRICE by POOL
```

```
F = 1.0334, num df = 485, denom df = 35, p-value = 0.9521
```

```
alternative hypothesis: true ratio of variances is not equal to 1
```

```
95 percent confidence interval:
```

```
0.5997214 1.6034567
```

```
sample estimates:
```

```
ratio of variances
```

```
1.033354
```

The ratio of variances is close to 1, and the p-value is high. So, we conclude that there is no evidence of different variances. Thus, the equal-variances T-test is justified.

4. Parallel boxplots

We can visualize the differences between the two samples by *parallel boxplots*. When we create a scatterplot with the first variable being categorical, R produces the following. The plot supports our findings about the means and variances.

```
> plot(POOL, SALES_PRICE)
```

